

Question # 24

 Revisit

which of the following is true for postman

Choose the best option

- ☒ postman is used to test web services.
- ☐ postman is a tool to test our database queries
- ☐ postman is a tool to test css and html
- ☐ none of the above

Clear Response

est, compile, verify in maven are called as

- ☐ phases
- ☐ goals
- ☐ lifecycle
- ☐ none of the above

Question # 20



In the following code , spring will create which objects

```
@RestController
```

```
public class UserController {
```

```
    @Autowired
```

```
    private UserService userService;
```

Choose the best option

- ☐ UserService object
- ☐ UserController object
- ☐ both UserController and UserService object.
- ☐ none of the above

Question # 20

[Revisit](#)

In the following code , spring will create which objects

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```

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Choose the best option

- ☒ UserService object
- ☐ UserController object
- ☐ both UserController and UserService object.
- ☐ none of the above

[Clear Response](#)

The function present in `CommandLineRunner` interface in spring is

- ☐ run
- ☐ runner
- ☐ runTask
- ☐ none of the above

the command to invoke some specific maven functionality in command line

- ☐ **maven**
- ☐ **build**
- ☐ **mvn**
- ☐ **none of the above**

Question # 18

 Revisit

In Hibernate when we give
`X x = session.get(X.class, ?)`. here ? is

Choose the best option

- ☐ tablename
- ☐ primary key value which is serializable
- ☐ databasename
- ☐ none of the above

which of the following is not a scope in jsp

- ☒ session
- ☐ request
- ☐ response
- ☐ page

Clear Response

```
@GetMapping("/startPage")  
public ModelAndView pocExample() {  
    ModelAndView modelAndView = new ModelAndView("x");  
    modelAndView.addObject("y", "z");  
    return modelAndView;  
}
```

Suppose x is a jsp, The preferred way to read the y value in the jsp
is using

- ☐ custom tags
- ☐ scriptlets
- ☐ directives
- ☐ actions



```
public class StockController {
```

```
    @autowired
```

```
    private StockService stockService;
```

```
    ..
```

```
}
```

What Annotation should we use on StockController

- ☐ @RestController
- ☐ @Presentation
- ☐ @Repository
- ☐ @Autowired

`String x=req.getParameter("y");` here y

- ☐ name of the parameter from the html
- ☐ id of the parameter from the html
- ☐ value of the parameter from the html
- ☐ none of the above

Question # 12

the configuration file used to do set proxy information,
in maven is called as

- ☐ pom.xml
- ☐ settings.xml
- ☐ proxy.xml
- ☐ none of the above

`@GetMapping`, `@PostMapping` are annotations used on

Choose the best option

- ☐ function
- ☐ class
- ☐ variables
- ☐ none of the above

Question # 34

to handle 404 in jee web applications of jee, we configure the file

- ☐ http.xml
- ☐ 404.xml
- ☐ web.xml
- ☐ none of the above

Question # 22

 Revisit

doGet and *doPost* methods return

Choose the best option

- ☐ int
- ☐ void
- ☐ String
- ☐ none of the above

Optional is a class present in

Question # 19

 Revisit

`<mapping = "" class = "?" />` ? here contains

Choose the best option

- ☐ name of the DAO class
- ☐ name of the Entity class
- ☐ name of the configuration file
- ☐ none of the above

Question # 23

 **Revisit**

To add a cookie , in the web applciation,we need to use object of

Choose the best option

- ☐ **HttpServletRequest**
- ☐ **HttpSession**
- ☐ **HttpServletResponse**
- ☐ **HttpCookie**

Question # 21

 [Revis](#)

@Repository
public interface UserRepository extends CrudRepository<User, Long> {
}
this class that implements this interface has to be implemented by

Choose the best option

- ☐ Programmer
- ☐ Spring
- ☐ depends on how you configure in application.properties
- ☐ none of the above

Question # 1

to contact nodejs server in react we use

Revisit

Choose the best option

- ☐ callAjax
- ☐ callServer
- ☒ callAsynchronousServer
- ☐ none of the above

Clear Response

`@autowired` can be used at

Question # 35

 Revisit

@autowired can be used at

Choose the best option

- ☒ function level
- ☐ variable level
- ☐ both function and variable level
- ☐ none of the above

Clear Response

Question # 32

 Revisit

`request.setAttribute(x,y);` here *x* must be of type

Choose the best option

- ☐ String
- ☐ Object
- ☐ Integer
- ☐ none of the above

Question # 36

 Revisit

which of the following annotation is used at function level

Choose the best option

- ☐ @component
- ☐ @Repository
- ☐ @Bean
- ☐ all the above

Question # 39



see the below code

```
@PostMapping("/api/employees/{id}")
```

```
public ResponseEntity<Boolean> updateStock(? Int id)
```

we are passing a stock object as part of Http request

to read this, annotation needs to be used is

Choose the best option

- ☐ @RequestParam
- ☐ @PathParam
- ☐ @PathVariable
- ☐ none of the above

Question # 33

 Revisit

`<jsp:useBean id=""x"" class=""Y"" scope=""request"" />`

this is in jsp is called as

 Choose the best option

- ☐ action
- ☐ directive
- ☐ declaration
- ☐ none of the above

to change tomcat port no in spring boot, property used in application.properties is.

- ☐ catalina.port
- ☐ server.port
- ☐ port.number
- ☐ none of the above

Question # 38

 Revisit

in JPA, to know whether particular entity with an id exists or not, we need to use the function.

Choose the best option

- ☐ isIdPresent
- ☐ existsById
- ☐ isEntityPresent
- ☐ none of the above

The method which is gets everytime a request is made in the servlet is

Choose the best option

- ☐ requestProcess
- ☐ service
- ☐ initialize
- ☐ none of the above

Question # 29



Revisit

to call a webservice from Spring, we need to use

Choose the best option

- ☐ CallWS
- ☐ RestTemplate
- ☐ CallWSviaJSON
- ☐ none of the above

to call a webservice from Spring, we need to use

@Table @Entity annotations **are used at**.



Question # 29

 Revisit

to call a webservice from Spring, we need to use

Choose the best option

- ☒ CallWS
- ☐ RestTemplate
- ☐ CallWSviaJSON
- ☐ none of the above

Clear Response

to call a webservice from Spring, we need to use

Question # 27

`<%@ taglib prefix="c" uri="http://java.sun.com/jsp/jstl/core"%>`
this line in jsp indicates usage of

- ☒ custom tags
- ☐ expressions
- ☐ scriptlets
- ☐ actions

Question # 28



Revisit

@GetMapping, @PostMapping are annotations used on

Choose the best option

- ☐ function
- ☐ class
- ☐ variables
- ☐ none of the above

Question # 26

 Revisit

String jdbcUrl = "jdbc:mysql://mysql.db.server:3306/xyz?useSSL=false&serverTimezone=UTC"
here 3306 is

Choose the best option

- ☐ password of database
- ☐ database name.
- ☐ portnumber where mysql is running
- ☐ none of the above

Which of the following ways below is correct to write a CSS?

Choose the best option

- ☐ p {color:red; text-align:center};
- ☐ p {color:red; text-align:center}
- ☐ p {color:red; text-align:center;}
- ☐ p (color:red;text-align:center,)



Which of the following is used to force putting a value in an input box?

Choose the best option

- ☐ placeholder
- ☐ forced
- ☐ mandatory
- ☐ required



Question # 6



Revisit

What does NaN stand for?

Choose the best option

- ☐ Name A Number
- ☐ Not A Name
- ☐ Not A Number
- ☐ Name A NoteScript

What is the correct JavaScript syntax to write "Hello World"?

Choose the best option

- ☒ `System.out.println("Hello World")`
- ☐ `print_r("Hello World")`
- ☐ `document.write("Hello World")`
- ☐ `response.write("Hello World")`

Clear Response

await returns which of the following

Revisit

Choose the best option

- ☐ promise
- ☐ value returned by promise
- ☐ a string.
- ☐ none of the above



render is a function present in

- ☐ Render
- ☐ React
- ☐ ReactDOM
- ☐ none of the above



Which of the following property is used to increase or decrease the size of a font?

- ☒ font-weight
- ☐ font-variant
- ☐ font
- ☐ font-size

Clear Response



Which of the class takes full width in Bootstrap?

Choose the best option

- ☐ Jumbotron
- ☐ navbar
- ☐ container
- ☐ container-fluid



```
const fs = require('fs');  
fs.unlink('./abc', (error) => {  
  if (error) throw error;  
  console.log('what happens');  
});  
In this code what is ""abc"" in the code.
```

Choose the best option

- ☐ filename
- ☐ databasename
- ☐ string
- ☐ none of the above



Question # 24

 Revisit

test.js contains the following code.

```
const express = require('express')  
const app = express()  
const port = 3000
```

```
app.post('/', (req, res) => {  
  res.send("Hello World!")  
})
```

```
app.listen(port, () => {  
  console.log('server running')  
})
```

When you run the program in nodejs and open the browser and type `http://localhost:3000` output will be

Choose the best option

- ☐ Hello World in browser, server running in nodejs console
- ☐ Hello World in browser only nothing in nodejs console
- ☐ server running alone and some error message in the browser
- ☐ none of the above



Next Question

render is a function present in

- ☒ Render
- ☐ React
- ☐ ReactDOM
- ☐ none of the above

Clear Response

A button in bootstrap can't be created using which tag?

Choose the best option

- ☐ button
- ☐ anchor
- ☐ input
- ☐ btn



Question # 5

closure is possible in javascript because

 Revisit

Choose the best option

- ☐ you can define a class within a class.
- ☐ You can write one variable inside another variable
- ☐ You can write one function inside another
- ☐ All the above



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1 of 40

>

All

Question # 1

Revisit

useSelector and useDispatch
are

Choose the best option

- ☐ React hooks
- ☐ ui events in react
- ☐ nodejs events
- ☐ None of the above

Question # 8



Which tag is used to specify that the browser is to fetch and use an external style sheet file?

Choose the best option

- ☐ link
- ☐ href
- ☐ src
- ☐ style

1 2 3 4 5 6 7 8 9 10

< 3 of 40 >

All

2

Question # 3

 Revisit

What is true about a callback function?

Choose the best option

- ☐ It is executed as soon as the calling function is executed
- ☐ It is executed as soon as the calling function has finished execution
- ☐ The callback function runs independent of the calling function
- ☐ The callback function doesn't return any data

react events are named using

- ☐ camelcase
- ☐ lowercase
- ☐ can be lowercase or camelcase.
- ☐ None of the above



export default X;
what is X here

 Revisit

Choose the best option

- ☒ name of the js file
- ☐ name of the object or class we created
- ☐ inbuilt react classname
- ☐ none of the above.





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2 of 40

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Question # 2

Revisit

Which sign does jQuery use as a shortcut for jQuery?

Choose the best option



\$

☐ %☐ #☐ &

Clear Response

Question # 10

 Revisit

for react to return multiple elements we can think of using

Choose the best option

- ☐ multiple
- ☐ many
- ☐ more
- ☐ Fragment



Which of the following operator is used to check greater than or equal to in MongoDB?

- ☐ \$gt
- ☐ \$gte
- ☐ gte
- ☐ any of the above

Question # 34

 Revisit

Table: EMP (EMPNO, ENAME, JOB, MGR, HIREDATE, SAL, COMM, DEPTNO)
Which of the following UPDATE command will set job to 'MANAGER' for 'SMITH'??

Choose the best option

- ☐ Update emp set job='MANAGER';
- ☐ Update emp set job='MANAGER' where ename = SMITH
- ☐ Update emp set job='MANAGER' where ename = 'SMITH'
- ☐ None of the above

Question # 35

 Revisit

What cannot be done on a view?

Choose the best option

☐ display

☐ filter

☐ index

☐ drop

Which type of join used, select * from emp e join dept d where e.deptno=d.deptno;

☒ left outer join

☐ inner join

☐ outer join

☐ none

Clear Response

Question # 33

 Revisit

Identify the correct operator

Select * from Students

where Country _____ ('India', 'Australia', 'Japan');

Choose the best option

☐ =

☐ LIKE

☐ IN

☐ All

Question # 7

 Revisit

to get redux component available in react we use

Choose the best option

- ☐ React redux
- ☐ React Provider
- ☐ Read ReduxSource
- ☐ none of the above

To delete a particular column in a relation the command used is:

- ☐ UPDATE
- ☐ DROP
- ☐ ALTER
- ☐ DELETE

 Revisit

Question # 40

Hierarchical model is also called

- Choose one
- ☐ Tree structure
 - ☐ Plex Structure
 - ☐ Normalize Structure
 - ☐ Table Structure

To delete a particular column in a relation the command used is:

☐ UPDATE

☐ DROP

☐ ALTER

☐ DELETE

Question # 25

 Revisit

Consider the following query :

`SELECT student, marks from table where marks > SELECT AVG(marks) from table` What function will be performed by the above query

 Choose the best option

- ☐ find students, and marks from table
- ☐ Query will show error
- ☐ find all the students, whose marks are greater than average marks
- ☐ Average marks of the students

Question # 32

Which of the following SQL statements deletes all rows in table called Sales?

 Revisit

Choose the best option

- ☐ DELETE Sales
- ☐ DELETE FROM Sales
- ☐ DELETE * FROM Sales
- ☐ DELETE ALL Sales

Question # 36



Revisit

Choose the best option

An ER diagram can express the overall _____ of a database graphically

- ☐ View structure
- ☐ logical structure
- ☐ model structure
- ☐ architechural structure

Question # 22

 Revisit

Choose the best option

Which of the following is an example of structured data

- ☒ CCTV generated data
- ☐ audio recording
- ☐ employee data
- ☐ None

Question # 2

 Revisit

Priority Queue with Heap gives both insertion and retrieval at

Choose the best

☒ $\log n$ ☐ n ☐ n^2 ☐ 2^n

Clear Response

Question # 37

 Revisit

Cutting down the height of recursion tree is the algorithm design technique called as

Choose the best option

- ☐ Backtracking
- ☐ Greedy approach
- ☐ Dynamic programming
- ☐ Divide and Conquer

Question # 19

 Revisit

Consider the join of a relation R with relation S . If R has m tuples and S has n tuples, then the maximum size of join is:

Choose the best option

- ☐ mn
- ☐ $m+n$
- ☐ $(m+n)/2$
- ☐ $2(m+n)$

Question # 24

 Revisit

A functional dependency is a relationship between or among

Choose the best option

- ☐ Entities
- ☐ Rows
- ☐ Attributes
- ☐ Tables

Question # 37

 Revisit

All aggregate functions except _____ ignore null values in their input collection.
Fill the blank with correct option

Choose the best option

- ☐ Count(attribute)
- ☐ Count(*)
- ☐ Avg
- ☐ Sum

Question # 16

 Revisit

If value of Left Subtree, value of node and value of Right Subtree are in strict order for all nodes in the tree, then we have a

Choose the best option

- ☐ AVL Tree
- ☐ Binary Search Tree
- ☐ Red Black Tree
- ☐ Hierarchical Tree

Question # 30

 Revisit

The time complexity of merging two sorted arrays is _____.

Choose the best option

- ☐ $O(n)$
- ☐ $O(n \log n)$
- ☐ $O(n^2)$
- ☐ $O(\log n)$

Question # 35

 Revisit

Solving the problem of a Maze is _____.

Choose the best option

- ☐ Brute force technique
- ☐ Dynamic programming
- ☐ Backtracking algorithm
- ☐ Greedy Approach

Question # 30

 Revisit

Choose the best option

Which keyword is used to create a variable in a stored procedure?

- ☐ UPDATE
- ☐ INSERT
- ☐ SELECT
- ☐ DECLARE

Question # 21

How can we specifies a row-level trigger?

 Revisit

Choose the best option

- ☐ Using ON ROW
- ☐ Using FOR EACH COL
- ☐ Using FOR EACH ROW
- ☐ Using OR ROW

Which of the following is correct CREATE INDEX Command?

- ☐ CREATE INDEX index_name ON database_name;
- ☐ CREATE INDEX index_name ON (index_columns);
- ☐ CREATE INDEX index_name ON table_name(index_columns);
- ☐ INSERT INDEX index_name ON table_name;

DATEDIFF function in mysql will display difference between 2 dates in terms of ———.

- ☐ number of months
- ☐ number of days
- ☐ number of years
- ☐ ALL of the above

Question # 28

Determine the maximum number of comparisons required to find the tomatoes with minimal and maximal weights among $2n$ tomatoes.

[Revisit](#)

Choose the best option

☐ $O(2n-1)$ ☐ $O(3n-2)$ ☐ $O(2n-2)$ ☐ $O(3n-1)$

Question # 20

 Revisit

Aggregate functions can be used in the select list or the _____ clause of a select statement or subquery. They cannot be used in a _____ clause.

Choose the best option

- ☐ Having, where
- ☐ Group by, having
- ☐ Group by, where
- ☐ Where, having

Question # 29

 Revisit

Find the minimal complexity for an algorithm that finds the rotten egg with minimal weight among a rack of good eggs.

Choose the best option

- ☐ $O(n \log n)$
- ☐ $O(n^2)$
- ☐ $O(\log n)$
- ☐ $O(n-1)$

Question # 31

 Revisit

A tree in which every node has either 0 or 2 children is called

Choose the best option

- ☐ Complete Binary Tree
- ☐ Strictly Binary Tree
- ☐ Unbalanced Binary Tree
- ☐ Binary Search Tree

Bubble Sort gives the best case when the array is

- ☐ Random
- ☐ Already sorted
- ☐ small
- ☐ big

Question # 38

 Revisit

Determine the functionality of the following code snippet:

```
public static String olympics(String tokyo) {  
    if (tokyo.length()==1)  
        return tokyo;  
    else  
        return tokyo.charAt(tokyo.length()-1) + olympics(tokyo.substring(0,tokyo.length()-1));  
}
```

Choose the best option

- ☐ Arranges the characters of the String in descending order
- ☐ Arranges the characters of the String in ascending order
- ☐ Finds whether the string is a palindrome
- ☐ Reverses a String

Question # 24

 Revisit

A sorted (ascending order) singly linked list has n nodes, determining the best case time complexity of inserting an element (new node) into the list such that the sorted order remains.

Choose the best option

- ☐ $O(\log n)$
- ☐ $O(1)$
- ☐ $O(n \log n)$
- ☐ $O(n)$

Question # 34

 Revisit

Determine the output of the following program:

```
class Ideone {  
    public static void main (String[] args) {  
        System.out.println(new Ideone().sarapatta(10));  
    }  
}
```

```
int sarapatta(int parambarai) {  
    if (parambarai==0 || parambarai==1)  
        return parambarai;  
    else  
        return (1*sarapatta(parambarai-2)+ 2*sarapatta(parambarai-1));  
    }  
}
```

Choose the best option

- ☐ 7823
- ☐ 3278
- ☐ 7832
- ☐ 2378

of degrees is

 Revisit

Choose the best option

- ☐ $2 \times \text{number of edges}$
- ☐ $\text{number of edges} / 2$
- ☐ $2 \times \text{number of vertices}$
- ☐ $\text{number of vertices} / 2$

Clear Response



Question # 36



Revisit

The collision resolution technique in Hashing where the next immediate vacant slot is looked for is called _____.

Choose the best option

- ☐ Quadratic Probing
- ☐ Rehashing
- ☐ Double Hashing
- ☐ Linear Probing

Question # 39

 Revisit

In tree traversal, root $\rightarrow$ left $\rightarrow$ right traversal style is called as

Choose the best option

- ☐ Post order traversal
- ☐ Depth First traversal
- ☐ Pre Order Traversal
- ☐ Breadth First Traversal

Question # 3

$f(n)$ represents time complexity of an algorithm, $g(n)$ is an arbitrary time complexity function. If $c_1g(n) \leq f(n) \leq c_2g(n)$, for every input size n (for all, $n > n_0$), where c_1, c_2 are constants ($c_1, c_2 > 0$), then

 Revisit

Choose the best option

- ☐ $f(n) = O(g(n))$
- ☐ $f(n) = \Omega(g(n))$
- ☐ $f(n) = \theta(g(n))$
- ☐ $f(n) = \Delta(g(n))$

 Revisit

ee, value of node and value of Right Subtree are in strict order for all nodes in the tree, then we



Choose the best option

- ☐ AVL Tree
- ☒ Binary Search Tree
- ☐ Red Black Tree
- ☐ Hierarchical Tree

Clear Response

Question # 20

 Revisit

Determine c_1 , c_2 , n_0 such that $c_1 \cdot n^2 \leq (n^2/2) - 3n \leq c_2 \cdot n^2$

Choose the best option

- ☐ $c_1=1/2, c_2=1/14, n_0=6$
- ☐ $c_1=1/2, c_2=1/14, n_0=7$
- ☐ $c_1=1/14, c_2=1/2, n_0=6$
- ☐ $c_1=1/14, c_2=1/2, n_0=7$

Question # 18

 Revisit

Determine the output of the following program:

```
class Chacha {  
    public static void main (String[] args) {  
        System.out.println(new Chacha().choudary(420,4));  
    }  
    int choudary(int n, int r) {  
        if(n>0)  
            return((n%r)+choudary(n/r,r));  
        else  
            return 0;  
    }  
}
```

Choose the best option

☒ 2☐ 3☐ 5☐ 6

Clear Response

Bubble Sort gives the best case when the array is

- ☐ Random
- ☐ Already sorted
- ☐ small
- ☒ big

Question # 11

 Revisit

In a Graph, the sum of degrees is

Choose the best option

- ☐ $2 \times \text{number of edges}$
- ☐ $\text{number of edges} / 2$
- ☐ $2 \times \text{number of vertices}$
- ☐ $\text{number of vertices} / 2$

Question 2 10

 Revisit

Determine the algorithm implemented in the following code snippet:

```
public int knockout(int[] data, int a, int b, int c) {
```

```
    int temp;
```

```
    if(b>c)
```

```
        return -1;
```

```
    temp=(b+c)/2;
```

```
    return (a == data[temp] ? temp : (a < data[temp] ? knockout(data,
```

Choose the best option

☒ Insertion Sort

☐ Binary Search

☐ Max Heapify

☐ Selection Sort

Clear Response

Question # 21

 Revisit

Choose the best option

Determine the output of the following program:

```
class LetsParty {  
    public static void main (String[] args) {  
        System.out.println(new LetsParty().mocktail(5));  
    }  
}
```

```
int mocktail(int missile) {  
    int rocket=0;  
    if(missile<=0)  
        return 1;  
    if(missile>3) {  
        rocket=missile;  
        return mocktail(missile-2)+2;  
    }  
    return mocktail(missile-1)+rocket;  
}
```

☐ 3

☐ 4

☐ 5

☐ 6

Question # 19

Priority Queue with Heap gives both insertion and retrieval at

Revisit

- ☐ $\log n$
- ☐ n
- ☐ n^2
- ☐ 2^n

Question # 18

 Revisit

Determine the output of the following program:

```
class Chacha {  
    public static void main (String[] args) {  
        System.out.println(new Chacha().choudary(420,4));  
    }  
    int choudary(int n, int r) {  
        if(n>0)  
            return((n%r)+choudary(n/r,r));  
        else  
            return 0;  
    }  
}
```

Choose the best option

- ☐ 2
- ☐ 3
- ☐ 5
- ☐ 6

Question # 17

 Revisit

All the internal nodes have both the children and all the leaves are at same level, then the tree is a

Choose the best option

- ☐ Complete Binary Tree
- ☐ Max Heap
- ☐ Min Heap
- ☐ AVL Tree

Question # 30

 Revisit

The time complexity of merging two sorted arrays is _____.

Choose the best option

- ☐ $O(n)$
- ☐ $O(n \log n)$
- ☐ $O(n^2)$
- ☐ $O(\log n)$

Question # 13

The 'push' operation is associated with _____ data structure.

 Revisit

Choose the best option

- ☐ Heap
- ☐ Stack
- ☐ Queue
- ☐ Array

The maximum number of edges in a simple graph of n vertices is

Choose the best option

☐ $(n(n-1))/2$

☐ $(n(n+1))/2$

☐ $(n-1)/2$

☐ $(n+1)/2$

Question # 33

 Revisit

Derived the Pre-order traversal of the tree whose Post order traversal and Inorder traversal are 9,8,12,20,15,10 and 8,9,10,12,15,20 respectively.

Choose the best option

- ☐ 8, 10, 9, 15, 12, 20
- ☐ 10, 8, 9, 15, 12, 20
- ☐ 8, 10, 9, 12, 15, 20
- ☐ 10, 8, 9, 12, 15, 20

The Stack is empty and a pop() operation is performed. What will the data structure report?

- ☒ Top
- ☐ Underflow
- ☐ Empty
- ☐ Overflow

[Clear Response](#)

Question # 13

The 'push' operation is associated with _____ data structure.

 Revisit

Choose the best option

- ☐ Heap
- ☐ Stack
- ☐ Queue
- ☐ Array

Question # 12

 Revisit

The First Come First Serve Process scheduling algorithm is a representation of the _____ data structure.

Choose the best option

- ☐ Heap
- ☐ Stack
- ☐ Queue
- ☐ Array

Question # 14

 Revisit

{'m','a','d'} are the initial contents of a stack. On this stack the following operations are performed: { push('a'), pop, push('m'), pop, push('i'), pop, pop}. What will be present content of the stack from bottom to top?

Choose the best option

- ☐ {'m'}
- ☐ {'m','a','d'}
- ☐ {'m','a','d','a'}
- ☐ {'m','a'}

Question # 8

 Revisit

Which of the following is the correct ascending order of the complexity expressions?

Choose the best option

- ☐ $n < 2n < n \log n < n^2 < 2^n < n!$
- ☐ $n < 2n < n \log n < n^2 < n! < 2^n$
- ☐ $n < n \log n < 2n < n^2 < 2^n < n!$
- ☐ $n < 2n < n \log n < 2^n < n^2 < n!$

Revise

Check with the solution

Question 10

Determine the algorithm implemented in the following code snippet:

```
public int knockout(int[] data, int a, int b, int c) {
```

```
    int temp;
```

```
    if (b > c)
```

```
        return -1;
```

```
    temp = (b + c) / 2;
```

```
    return (a == data[temp] ? temp : (a < data[temp] ? knockout(data,
```

```
        a, temp + 1, c)) :
```

```
        knockout(data,
```

☐ Insertion Sort

☐ Binary Search

☐ Max Heapify

☐ Selection Sort

Question # 1

Determine the sorting algorithm implemented in the following code snippet.

[Revisit](#)**Choose the best option**

```
public void tobesorted(int[] data){  
    int tmp;  
    int i, j;  
    for(i = 1; i < data.length; i++){  
        tmp = data[i];  
        for(j=i; j>0 && tmp < data[j-1]; j--)  
            data[j] = data[j-1];  
        data[j] = tmp;  
    }  
}
```

- ☐ Insertion Sort
- ☐ Selection Sort
- ☐ Bubble Sort
- ☐ Shell Sort

Question # 7

 Revisit

The Stack is empty and a `pop()` operation is performed. What will the data structure report?

Choose the best option

- ☐ Top
- ☐ Underflow
- ☐ Empty
- ☐ Overflow

Question # 5

Determine the postfix notation for the infix notation: $(a+b) * (c+d)$

☐ $abcd++^*$

☐ $ab+cd^*+$

☐ $abcd+^*+$

☐ $ab+cd+^*$

The Fractional Knapsack problem is an example for _____.

- ☐ Brute force technique
- ☐ Dynamic programming
- ☐ Backtracking algorithm
- ☐ Greedy Approach

Question # 6

Determine the output of the following program:

```
class Chachi {  
    public static void main (String[] args) {  
        new Chachi().foodie(888);  
    }  
}
```

```
void foodie(int noodles) {  
    if(noodles<=1) {  
        System.out.print(noodles);  
    } else {  
        foodie(noodles/2);  
        System.out.print(noodles%2);  
    }  
}
```

 Revisit

Choose the best option

- ☐ 1110111000
- ☐ 1101111000
- ☐ 1100111000
- ☐ 1101101000

Determine the algorithm implemented in the following code snippet:

```
public int knockout(int[] data, int a, int b, int c) {  
    int temp;  
    if(b>c)  
        return -1;  
    temp=(b+c)/2;  
    return (a == data[temp] ? temp : (a < data[temp] ? knockout(data, a, b, temp-1) : knockout(data,  
a, temp+1, c)));  
}
```

Choose the best option

- ☐ Insertion Sort
- ☐ Binary Search
- ☐ Max Heapify
- ☐ Selection Sort

Question # 2

Which of the following is not a linear data structure?

[Revisit](#)

Choose the best option

- ☐ Array
- ☐ Linked List
- ☐ Tree
- ☐ Vector

An element inside an object can't be accessed using?

Choose the best option

- ☐ `object.element`
- ☐ `object["element"]`
- ☐ `object::element`
- ☐ `JSON.stringify(object["element"])`

Question # 32

 Revisit

Determine n_0 where the expression $(6n^2)$ becomes greater than $(100n+300)$ for $n>0$.

Choose the best option

- ☐ 20
- ☐ 21
- ☐ 22
- ☐ 23

Which of the following is not a state for a Promise?

Choose the best option

- ☐ pending
- ☐ fulfilled
- ☐ rejected
- ☐ error

Question # 32



Revisit

when the component is loaded for the first time in react the event is called.

Choose the best option

- ☐ componentDidMount
- ☐ ComponentDidstart
- ☐ ComponentDidinit
- ☐ none of the above.

In react, you see some line as follows

```
const element = <header input=""hello"" />;
```

To read hello given in above line, in header component we need to use

 Revisit

Choose the best option

- ☐ attribute
- ☐ props
- ☐ header.input
- ☐ none of the above

Question # 1

 Revisit

Determine the sorting algorithm implemented in the following code snippet:

```
public void tobesorted(int[] data){  
    int tmp;  
    int i, j;  
    for(i = 1; i < data.length; i++){  
        tmp = data[i];  
        for(j=i; j>0 && tmp < data[j-1]; j--)  
            data[j] = data[j-1];  
        data[j] = tmp;  
    }  
}
```

Choose the best option

- ☐ Insertion Sort
- ☐ Selection Sort
- ☐ Bubble Sort
- ☐ Shell Sort

Which HTML attribute is used to define inline styles?

Choose the best option

- ☐ style
- ☐ inline
- ☐ inline-style
- ☐ inline-attr



of the following property is used to increase or decrease the size of a font?



- ☒ font-weight
- ☒ font-variant
- ☒ font
- ☐ font-size

Clear Response

The _____ selector is used to specify a style for a single, unique element.

Revisit

Choose the best option

☐ id

☐ class

☐ unique

☐ select

Question # 4

 Revisit

Which of the following sorting algorithms has the best, average and worst case time complexities as $O(n^2)$?

Choose the best option

- ☐ Merge Sort
- ☐ Heap Sort
- ☐ Insertion Sort
- ☐ Selection Sort

to contact nodejs server in react we use

Choose the best option

- ☐ callAjax
- ☐ callServer
- ☐ callAsynchronousServer
- ☐ none of the above

function used to assign http headers in node js http module is

Choose the best option

- ☐ setHttpHeader
- ☐ acceptHttpHeader
- ☐ putHttpHeader
- ☐ writeHead

HttpStatus code 201 indicates

Choose the best option

- ☐ method is not supported on the url
- ☐ resource is not found
- ☐ Internal Server Error
- ☐ none of the above



Question # 2

Revisit

Which size class is not available in Bootstrap?

Choose the best option

- ☒ md
- ☐ lg
- ☐ xs
- ☐ xl

Clear Response



-
-
-
-



or



Which HTML attribute is used to define inline styles?

Choose the best option

- ☐ style
- ☐ inline
- ☐ inline-style
- ☐ inline-attr



The _____ selector is used to specify a style for a single, unique element.

Choose the best option

- ☒ id
- ☐ class
- ☐ unique
- ☐ select

Clear Response

1000
1000
1000
1000



When install node package it gets created in the folder called.

 Revisit

Choose the best option

- ☐ node_packages
- ☐ lib
- ☐ node_modules
- ☐ None of the above



Question # 21

 Revisit

Which one of these technologies is NOT used in AJAX?

Choose the best option

- ☐ DHTML
- ☐ FLASH
- ☐ DOM
- ☐ XML

Question # 26

Revisit

which of the following statements are true in react

- ☐ state and props both are immutable
- ☐ state can be used within the component only
- ☐ state is immutable
- ☐ None of the above



1. The first step in the process is to identify the problem.



- The second step is to analyze the problem.
- The third step is to develop a plan.
- The fourth step is to implement the plan.



1. The first step in the process is to identify the problem.

10

Which jQuery function is used to send ajax request using GET method?

 Choose the best option

- ☐ \$.ajax()
- ☐ \$.get()
- ☐ \$_GET()
- ☐ \$_AJAX()

data is not handled by React is called as

Choose the best option

- ☐ React Component
- ☐ UI Component
- ☐ UnControlled Component
- ☐ None of the above

Term document in MongoDB is equivalent to _____ in MySQL

- ☐ Row
- ☐ table
- ☐ attribute
- ☐ Greater Entity Count (GEC)

What should be the root tag in any HTML document?

Choose the best option

- ☐ root
- ☐ html
- ☐ doctype
- ☐ head



Calculate the average waiting time using shortest job first (without preemption) where we have 4 processes with arrival and burst Time.

Process	-	P1	P2	P3	P4
Arrival time	-	2	1	0	5
Burst time	-	4	3	2	8

- ☐ 2
- ☐ 1.7
- ☐ 4
- ☐ 4.2

Which of the following program will bind the 'relocatable addresses' in an executable file to the 'absolute addresses' at the time of execution

- ☐ Assembler
- ☐ Process Scheduler
- ☐ Loader
- ☐ Translation Lookaside Buffer (TLB)

A process that has completed its execution but still continues to exist as a process is called

- ☐ Orphan process
- ☐ Zombie process
- ☐ daemon process
- ☐ Child process

Question # 26

Git command to compare two specified branches

 Revisit

Choose the best option

- ☒ git diff
- ☐ git merge
- ☐ git blame -L
- ☐ git push --tags

Choose an example of a UNIX system call from the following

☐ printf

☐ write

☐ main

☐ fopen

A BREAK statement inside a Loop like WHILE, FOR, DO WHILE and Enhanced-FOR causes the program execution ____ Loop.

- ☐ Exit
- ☐ Continuation with next iteration
- ☐ Never exit
- ☐ None of the above

It refers to the degree to which a module's instructions are functionally related

- ☐ Functional testing
- ☐ Coupling
- ☐ Beta testing
- ☐ Cohesion

What is the output of the code snippet with the ternary operator?

```
int p=5;  
System.out.print("Hello ");  
(p<5)?5:6;
```

- ☐ Hello 5
- ☐ Hello 6
- ☐ Hello
- ☐ Compiler error

- ☐ System design life cycle
- ☐ Software development life cycle
- ☐ System development life cycle
- ☐ Software requirement life cycle

When a process calls fork what is the return value the process should receive on success?

- ☐ 0
- ☐ Child Process Id
- ☐ Greater than 0
- ☐ parent process Id

The PID of the `init` process on UNIX systems is

- ☐ 10
- ☐ 0
- ☐ 1
- ☐ 100

What happens to the Second operand/expression if the first operand is FALSE with a Short Circuit AND (&&) operator?

- ☒ Second operand/expression is evaluated and AND is applied.
- ☐ Evaluation of Second operand/expression is skipped.
- ☐ The compiler starts taking more memory
- ☐ The compiler starts taking more CPU power.

Clear Response

Which of the following program will bind the relocatable addresses in an executable file to the absolute addresses at the time of execution?

Revisit

Choose the best option

- ☐ Assembler
- ☐ Process Scheduler
- ☐ Loader
- ☐ Translation Lookaside Buffer (TLB)

Q. 22

Message Passing allows

Revisit

Choose the best option

- ☐ process to communicate without using shared data
- ☐ process to communicate with using shared data
- ☐ threads to communicate without using shared data
- ☐ threads to communicate with using shared data

Question # 21

What is the output of the Java code snippet?

```
int k=20;
```

```
if(k)
```

```
{
```

```
    System.out.println("YES");
```

```
}
```

```
else
```

```
{
```

```
    System.out.println("NO");
```

```
}
```

 Revisit

Choose the best option

☐ NO

☐ YES

☐ Compiler error

☐ None of the above

when we use `PreparedStatement.executeUpdate()`;

when we are doing an update, if update fails due to where condition. we get

- ☐ return int as zero
- ☐ return int as 1
- ☐ it throws SQLException
- ☐ none of the above.

Question # 27

 Revisit

Which procedure parameter enables the caller to pass in a value and get back a value?

Choose the best option

- ☐ IN
- ☐ OUT
- ☐ INOUT
- ☐ GETINOUT

Question 100 (text question)

- ☐ More than one program/job/process can reside in the main memory at the same point of time
- ☐ More than one task/program/job can reside in the same CPU at the same point of time
- ☐ a computer system in which multiple terminals connect to a host computer that handles processing tasks
- ☐ the active tasks are swapped from active CPU and kept aside and another task is brought in CPU for a very small fraction of time and swapped back to the earlier task again



The kernel of the operating system remains in the primary memory because _____.

- Without the best option
- ☒ It is mostly called (used)
 - ☐ It manages all interrupt calls
 - ☐ It controls all operations in process
 - ☐ It is low level

What is a Widening Type Conversion in Java?

- ☒ Conversion of data from higher data type to lower data type
- ☐ Conversion of data from lower data type to higher data type
- ☐ Conversion of data from any data type to any data type
- ☐ None of the above

`session.save(entityobject)` in hibernate returns

- ☐ serializable instance of the primary key
- ☐ void
- ☐ boolean
- ☐ none of the above

- ☐ parent of parent process
- ☐ parent process
- ☐ init process
- ☐ swapper process

Question # 29

 Revisit

The string function that returns the index of the first occurrence of substring is _____

Choose the best option

- ☐ INSERT()
- ☐ INSTR()
- ☐ INSTRING()
- ☐ INFSTR()

Question # 11

 Revisit

which of the following is not a scope in jsp

Choose the best option

- ☐ session
- ☐ request
- ☐ response
- ☐ page

Question # 6

to test a spring MVC Rest web application we can use a class called

 Revisit

Choose the best option

- ☐ TestRestTemplate
- ☐ TestTemplate
- ☐ MockRestTemplate
- ☐ none of the above

Question #4

 Revisit

Choose the best option

Which of the following is not the approach to handling deadlocks

- ☐ Deadlock Prevention
- ☐ Deadlock Avoidance
- ☐ Detect and Recover
- ☐ Virtual Memory

Question # 4



Revisit

```
<bean id="x" class="com.service.ServiceImplementation.">  
  <property name="port" value="23">  
  </property>
```

this will call a function called

Choose the best option

- ☐ setPort in ServiceImplementation
- ☐ port in ServiceImplementation.
- ☐ getPort in ServiceImplementation
- ☐ none of the above

Choose the best option

- ☐ E Commerce frame work
- ☐ Integrated Development Environment
- ☐ It is widely used in software development and software testing
- ☐ Compiler

To ensure spring creates a new object everytime we ask for it.
the scope needs to be

- ☐ singleton
- ☒ prototype
- ☐ clone
- ☐ copy

`public class Demo extends GenericServlet { }`
which method we have to override

Choose the best option

- ☐ doGet
- ☐ doPost
- ☐ service
- ☐ none of the above.

In Hibernate, we use
`Transaction t = ?.beginTransaction();`, here ? is of type

- ☐ Session
- ☐ HibernateSession
- ☐ TransactionFactory
- ☐ none of the above

which of the following statements are true in react

- ☐ state and props both are immutable
- ☒ state can be used within the component only
- ☐ state is immutable
- ☐ None of the above

Clear Response

Question # 3



Revisit

`statement.executeQuery(query)` returns an object of

Choose the best option

- ☐ Query
- ☐ Selectable
- ☐ PreparedStatement
- ☐ none of the above

What is the output of the below Java program with arrays?

```
String[] colors = {"RED", "YELLOW", "WHITE"};
```

```
System.out.print(colors[2]);
```

Choose the best option

- ☐ RED
- ☐ YELLOW
- ☐ WHITE
- ☐ Compiler error

Standard for Software Lifecycle

- ☐ 1070
- ☐ X.509
- ☐ 802.x
- ☐ 1074

What happens to the Second operand/expression if the first operand is FALSE with a Short Circuit AND (&&) operator?

- ☐ Second operand/expression is evaluated and AND is applied.
- ☐ Evaluation of Second operand/expression is skipped.
- ☐ The compiler starts taking more memory.
- ☐ The compiler starts taking more CPU power.

@Bean annotation must be used on functions

Compaction technique is possible in

- ☐ paging
- ☐ scheduling
- ☐ external fragmentation
- ☐ internal fragmentation

What is the output of the following script?

- ☐ if [\$reply = "Y"]; then
- ☐ if [\$reply = "Y"], then
- ☐ if test \$reply = "Y"; then
- ☐ if (\$reply = "Y"); then

Which of the following function will remove collection in MongoDB?

- ☐ `db.drop()`
- ☐ `db.dropCollection()`
- ☐ `db.remove()`
- ☐ `db.collection.drop()`